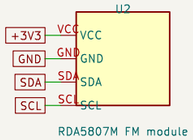
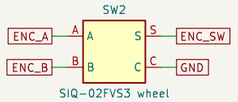
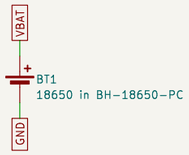
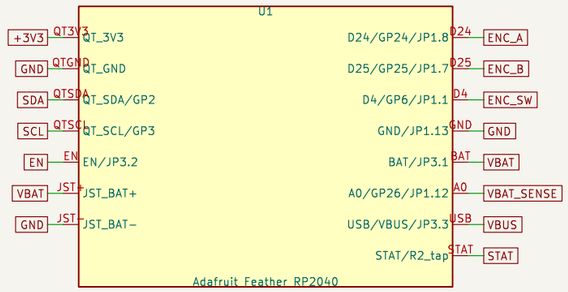
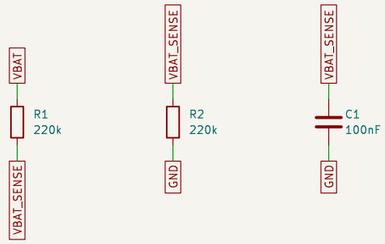
OpenALD Feather RP2040 v1.0 — hand-wiring schematic

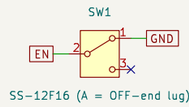


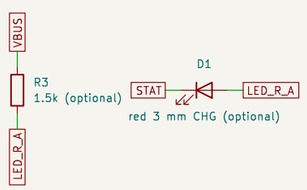


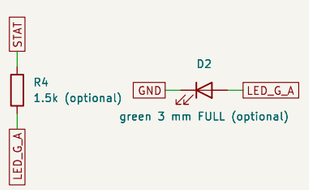












OpenALD Feather RP2040 v1.0: every net below is a hand-soldered wire (no PCB). NOT built or tested on hardware.

GND: Feather JP1.13 is the only GND pin -> heat-shrunk splice to SW2 C, R2, C1, D2 K. SW1 OFF lug -> splice on the battery black lead (JST- = GND).

SW1: knob at the USB end = OFF: common (EN) connects to the OFF-end lug (GND) -> 3.3 V regulator off. Check which lug with a meter.

U1 JST+ and BAT are the same node on the Feather (VBAT); 3V3 (STEMMA QT) is switched by EN.

R3/D1/R4/D2 optional: STAT low (charging) -> red, STAT high (full) -> green, no USB -> both off. STAT = fine wire to the STAT end of Feather R2.

Firmware: firmware/openald_feather (GP24 ENC_A, GP25 ENC_B, GP6 ENC_SW, GP26 VBAT_SENSE = VBAT/2, I2C1 GP2/GP3, NeoPixel GP16).

Not hardware-validated
Generated by scripts/make_schematic.py — same nets as docs/wiring_diagram.svg

Sheet: /
File: openald-feather-wiring.kicad_sch

Title: OpenALD Feather RP2040 v1.0 — wiring

Size: A3	Date: 2026-11-12	**Rev: 1.0**
KiCad E.D.A. 9.0.2		Id: 1/1